EXPLORATORY DATA ANALYSIS

Student Marks Dataset | Mini Project Report

1. Objective

To inspect the quality of student study-hour and exam-mark data, clean duplicate and missing records, summarize the numeric variables, and visualize how study hours relate to exam marks.

2. Dataset Overview

The provided CSV contains **103 rows** and **3 columns**: Student_ID (identifier), Study_Hours (numeric), and Exam_Marks (numeric). The dataset source available for this report is the user-provided *student_marks.csv*; no external source URL was included.

Data-quality check	Observed result
Initial rows / columns	103 / 3
Missing Study_Hours	4
Missing Exam_Marks	3
Duplicate rows	3
Rows after duplicate removal and IQR filtering	97

3. Data Cleaning

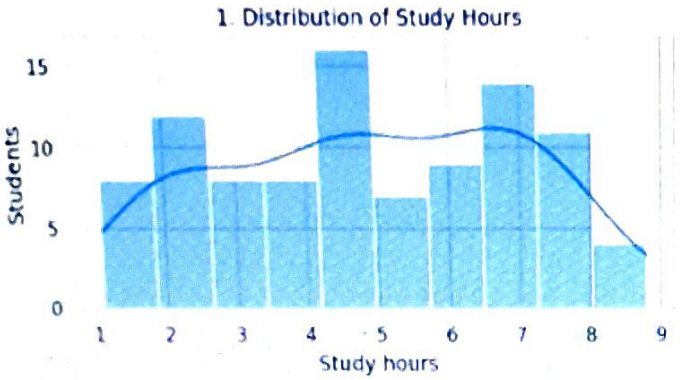
Exact duplicate rows were removed (3). Missing Study_Hours and Exam_Marks values were filled with the median of each column. The notebook then applied the IQR rule to Study_Hours: Q1 = 3.20, Q3 = 6.65, IQR = 3.45, with bounds -1.98 to 11.82. It removed 3 rows outside those bounds, leaving 97 records. No missing values remain in the two numeric fields after median imputation.

4. Descriptive Analysis

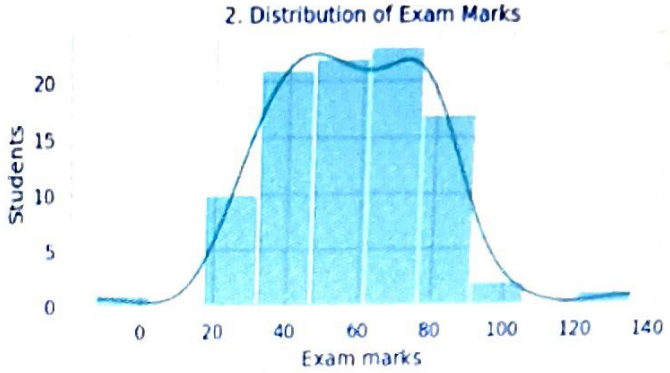
Statistic	Study_Hours	Exam_Marks
Count	97	97
Mean	4.91	57.61
Median	4.60	56.00
Std. deviation	2.11	21.46
Minimum	1.10	-12.00

5. Visual EDA and Observations

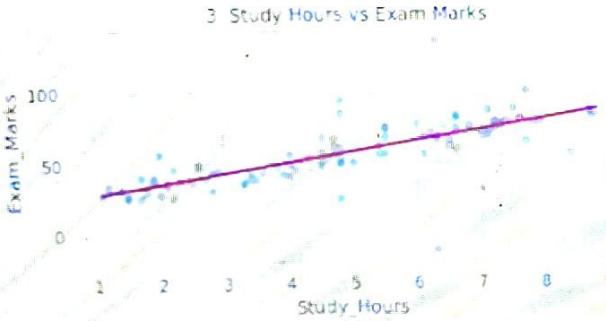
The seven charts below examine distributions, outliers, the study-hours/marks relationship, correlation, and average marks by study-hour group. The Pearson correlation after cleaning is **r = 0.748**, indicating a positive linear association in this dataset.

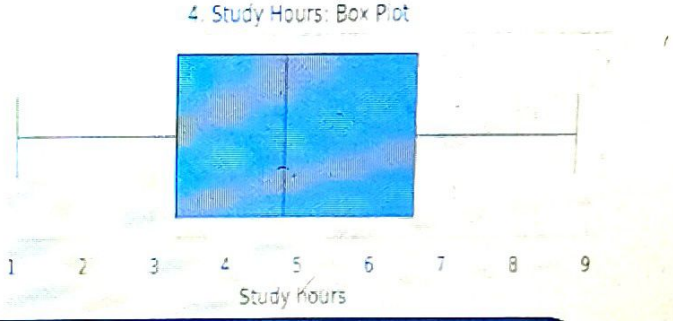


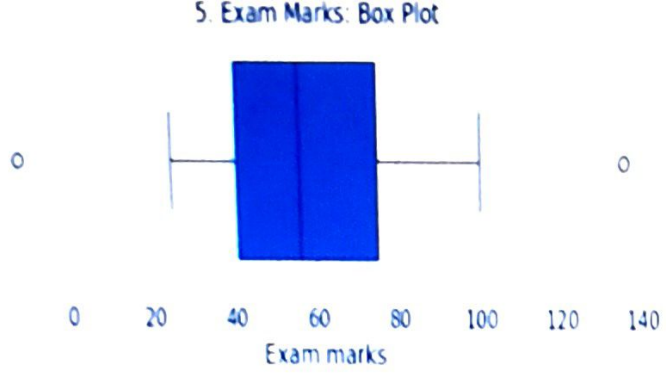
Study-hour distribution



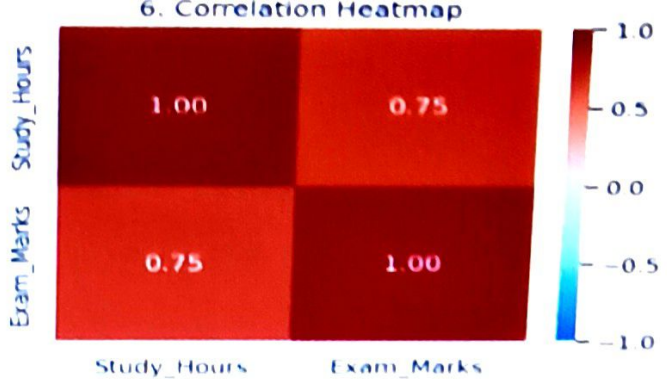
Exam-mark distribution



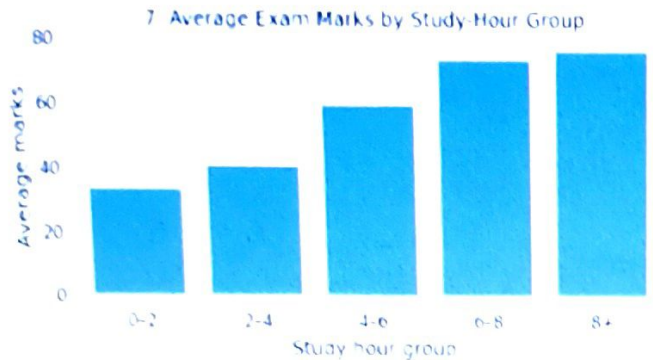




Exam-mark box plot



Correlation heatmap



Mean marks by study-hour group

6. Evidence-Based Findings

- After cleaning, 97 rows remain from the original 103 records
- The average study time is 4.91 hours; the median is 4.80 hours
- The average exam mark is 57.61; the median is 56.00
- Study hours and exam marks have a positive correlation ($r = 0.748$), this is an association and does not prove causation
- Average marks by study-hour band are: 0–2 hours: 33.75; 2–4 hours: 41.00; 4–6 hours: 59.88; 6–8 hours: 73.85; 8+ hours: 76.50

7. Limitations and Conclusion

This dataset has only three columns and no context about exam maximum, subject, attendance, prior achievement, or other factors that may affect marks. The source notebook treats outliers only in Study_Hours; exam marks are not range-validated. The uploaded CSV contains exam values outside a typical 0–100 scale (including negative values and values above 100), so these should be checked with the dataset owner before drawing strong conclusions. Overall, the EDA workflow demonstrates data-quality checks, median imputation, IQR filtering, descriptive statistics, and seven visualizations. Correlation should not be interpreted as proof that study time alone causes higher marks.